

High-density Lipoprotein Cholesterol (HDL-C)

Test Details

METHODOLOGY

Enzymatic

RESULT TURNAROUND TIME

Within 1 day

Turnaround time is defined as the usual number of days from the date of pickup of a specimen for testing to when the result is released to the ordering provider. In some cases, additional time should be allowed for additional confirmatory or additional reflex tests. Testing schedules may vary.

Use

Monitoring of HDL-cholesterol in serum is of clinical importance since an inverse correlation exists between serum HDL-cholesterol concentrations and the risk of atherosclerotic disease. Elevated HDL-cholesterol concentrations are protective against coronary heart disease, while reduced HDL-cholesterol concentrations, particularly in conjunction with elevated triglycerides, increase the cardiovascular risk.^{2,3} Strategies have emerged for treating cardiovascular disease by increasing HDL-cholesterol levels.^{4,5}

Custom Additional Information

Total cholesterol and triglycerides are required as well for determination of lipid risk factors for coronary artery disease. These tests with HDL-C and LDL-C are the usual lipid profile. HDL-C is especially apt to be low in male subjects who are obese and sedentary, in those who smoke cigarettes, and in those who have diabetes mellitus. Uremia is also associated with lower HDL-C. Exercise, appropriate diet and moderate ethanol intake increase HDL-C.

HDL-C is useful with cholesterol in forecasting protection against coronary artery disease in the industrialized countries, possible because of ingestion of high fat diets.

Those at least risk for development of coronary arterial disease have low cholesterol, low triglycerides, and high HDL-C.

Factors contributing to decreased HDL-C include:

- Genetic factors: primary hypoalphalipoproteinemia⁶
- Cigarette smoking⁷
- Obesity⁷
- Hypertriglyceridemia⁷
- Lack of exercise
- Steroids – androgens, progestogens, anabolic
- Thiazides
- Beta-adrenergic blockers
- Probucol

- Neomycin

Specimen Requirements

Specimen

Serum (preferred) **or** plasma

Volume

1 mL

Minimum Volume

0.7 mL (**Note:** This volume does **not** allow for repeat testing.)

Container

Red-top tube, gel-barrier tube, **or** green-top (lithium heparin) tube. Do **not** use oxalate, EDTA, or citrate plasma.

Collection Instructions

Separate serum or plasma from cells within 45 minutes of collection.

Stability Requirements

Temperature	Period
Room temperature	3 days
Refrigerated	14 days
Frozen	14 days
Freeze/thaw cycles	Stable x2

Reference Range

>39 mg/dL

Storage Instructions

Maintain specimen at room temperature.¹

Patient Preparation

Patient should be on a normal diet and maintain a stable weight for a week prior to testing. Any drugs should be discontinued for three to four weeks if possible. Test should not be performed until three months after a myocardial infarction or similar traumatic episode, such as severe infection or inflammation. Fasting is not necessary.

Causes for Rejection

Improper labeling

Footnotes

1. LabCorp internal studies.
2. Nauck M, Wiebe D, Warnick G. Measurement of high-density-lipoprotein. In: Rifai N, Warnick CR, Dominiczak MH, eds. *Handbook of Lipoprotein Testing*. 2nd ed. Washington, DC: AACC Press; 2001: 221-224.
3. Dominiczak M, McNamara J. The system of cardiovascular prevention. In: Rifai N, Warnick CR, Dominiczak MH, eds. *Handbook of Lipoprotein Testing* 2nd ed. Washington, DC: AACC Press; 2001:103-125.
4. Linsel-Nitschke P, Tall AR. HDL as a target in the treatment of atherosclerotic cardiovascular disease. *Nat Rev Drug Discov*. 2005 Mar; 4(3):193-205. PubMed 15738977 (<http://www.ncbi.nlm.nih.gov/pubmed/15738977>)
5. Ng DS. Treating low HDL—from bench to bedside. *Clin Biochem*. 2004 Aug; 37(8):649-659. PubMed 15302606 (<http://www.ncbi.nlm.nih.gov/pubmed/15302606>)
6. Third Report of the National Cholesterol Education Program (NCEP) Expert Panel on Detection, Evaluation, and Treatment of High Blood

Cholesterol in Adults (Adult Treatment Panel III) Final Report. *Circulation*. 2002 Dec 17; 106(25):3143-3421. PubMed 12485966

(<http://www.ncbi.nlm.nih.gov/pubmed/12485966>)

7. Frohlich JJ, Pritchard PH. The clinical significance of serum high density lipoproteins. *Clin Biochem*. 1989 Dec; 22(16):417-423. PubMed 2692872

(<http://www.ncbi.nlm.nih.gov/pubmed/2692872>)